

**FIFTH SEMESTER U.G. DEGREE (SPECIAL) EXAMINATION
NOVEMBER 2020**

(CUCBCSS—UG)

Mathematics

MAT 5B 08—DIFFERENTIAL EQUATIONS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 5B 08—DIFFERENTIAL EQUATIONS

(Multiple Choice Questions for SDE Candidates)

1. Which of the following is a linear differential equation ?

(A) $y'' + (y')^2 = \sin x.$

(B) $(y'')^2 + 3y = e^x.$

(C) $y'' + 3y' + y = 0.$

(D) $(y'')^2 + (y')^3 + e^x = 0.$

2. Which of the following is a separable differential equation ?

(A) $\frac{dy}{dx} = \frac{x^2}{1 - y^2}.$

(B) $\frac{dy}{dx} = \frac{x + y}{x - y}.$

(C) $\frac{dy}{dx} + (\sin x) y = e^x.$

(D) $\left(\frac{dy}{dx}\right)^2 + (\sin x) y = 0.$

3. The general solution of the differential equation $2x(3x + y - ye^{-x^2})dx + (x^2 + 3y^2 + e^{-x^2})dy = 0$ is :

(A) $x^2y + ye^{-x^2} + 2x^3 + y^3 = C.$

(B) $x^2y^2 + ye^{x^2} + 2x + y^2 = C.$

(C) $xy + ye^{-x^2} + y^2 = C.$

(D) $xy^2 + y + 2x^3e^{-x^2} + y^3 = C.$

4. Which of the following is an initial value problem ?

(A) $y' + y = 0, y(0) = y'(0) = 0.$

(B) $y' + y = 0, y(0) = y(1) = 0.$

(C) $y'' + y = 0, y(0) = 0, y(1) = 1..$

(D) $y''' + y = 0, y(0) = 0, y(2) = 4.$

5. Which of the following is a boundary value problem :

(A) $y' + y = 0, y(0) = 1, y'(0) = 0.$

(B) $y'' + 5y = 0, y(0) = 1, y'(0) = 3.$

(C) $x^2y'' + xy' + y = 0, y(0) = 0, y(1) = 2.$

(D) $y''' + y'' + y = 0, y(0) = y'(0) = y''(0) = 0.$

6. The general solution of the differential equation $y' = \cos x$ is :

(A) $y = \sin x.$

(B) $y = \cos x.$

(C) $y = C \sin x.$

(D) $y = \sin(x) + C.$

7. If $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is a function of x only, then an integrating factor of $Mdx + Ndy = 0$ is :

(A) $\mu(x) = \exp \left[\int \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dx \right].$

(B) $\mu(x) = \exp \left[\int \frac{1}{N} \left(\frac{\partial M}{\partial y} + \frac{\partial N}{\partial x} \right) dx \right].$

(C) $\mu(x) = \int \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dx.$

(D) $\mu(x) = \int \frac{1}{N} \left(\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y} \right) dx.$

8. If $\frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$ is a function of y only, then an integrating factor of the differential equation

$Mdx + Ndy = 0$ is :

(A) $\mu(x) = \exp \left[\int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy \right].$

(B) $\mu(x) = \exp \left[\int \frac{1}{M} \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y} \right) dy \right].$

(C) $\mu(x) = \int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy.$

(D) $\mu(x) = \int \frac{1}{M} \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y} \right) dy.$

9. An integrating factor of the differential equation $\frac{dx}{dy} + P(x) = Q(x)$ is :

(A) $e^{\int p dx}.$

(B) $e^{-\int p dx}.$

(C) $e^{\int p^2 dx}.$

(D) $e^{\int (P+Q) dx}.$

10. A mathematical model of an object falling in the atmosphere near the surface of earth is given by :

(A) $m \frac{dv}{dt} = mg - rv.$

(B) $m \frac{d^2v}{dt^2} = mg - rv.$

(C) $\frac{dv}{dt} = mg.$

(D) None of these.

11. The differential equation $y'' - 5y' + 6y = 0$ has :

(A) Two linearly independent solutions.

(B) Three linearly independent solutions.

(C) Four linearly independent solutions.

(D) Infinite number of linearly independent solution.

12. The general solution of the differential equation $(D^2 - 4D + 4)y = 0$ is :

(A) $(c_0 + c_1x)e^{2x}.$

(B) $(c_0 - c_1x)e^{2x}.$

(C) $c_1e^x c_2e^{-2x}.$

(D) $c_1e^{2x} + c_2e^{2x}.$

13. The characteristic roots of the differential equation $(D^2 - 2D)y = 4x^2 + 2x + 3$ are :

(A) $\lambda = 0, \lambda = -2.$

(B) $\lambda = 1, \lambda = 3.$

(C) $\lambda = 0, \lambda = 2.$

(D) $\lambda = 1, \lambda = -2.$

14. The Laplace transform of the unit step function $u_{(t-a)}$ is :

(A) $e^{-as}.$

(B) $e^{-as}/s.$

(C) $e^{as}/s.$

(D) $e^{-as}/s^2.$

15. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{f(at)\} =$

(A) $\frac{1}{a} F(s/a).$

(B) $F(s/a).$

(C) $F(a/s).$

(D) $F(s).$

16. $\int_0^\infty \frac{\sin t}{t} dt =$

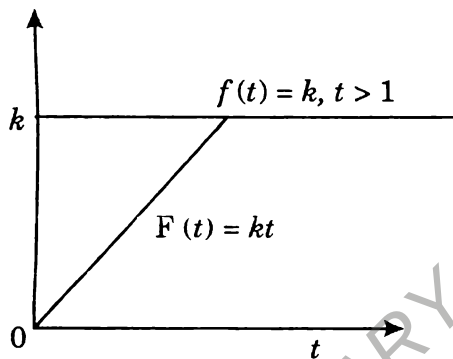
(A) $\frac{\pi}{4}.$

(B) $\frac{\pi}{8}.$

(C) $\frac{\pi}{2}.$

(D) None of these.

17. The Laplace transform of the function whose graph shown below is :



(A) $\frac{k}{s^2} (1 - e^{-s}).$

(B) $\frac{k}{s} (1 - e^{-s}).$

(C) $\frac{1}{s} (1 - e^{-s}).$

(D) None of these.

18. $\mathcal{L}\{\sin \hat{a}t\} =$

(A) $\frac{a}{s^2 - a^2}.$

(B) $\frac{a}{s^2 + a^2}.$

(C) $\frac{s}{s^2 - a^2}.$

(D) $\frac{s}{s^2 + a^2}.$

19. $\mathcal{L}\{t^n\} =$

(A) $\frac{n!}{s^n}.$

(B) $\frac{(n+1)!}{s^n}.$

(C) $\frac{n!}{s^n + 1}.$

(D) $\frac{1}{s^n}.$

20. The solution to the problem :

$$\alpha^2 u_{xx} = u_{tt}, 0 \leq x \leq L$$

$$\begin{cases} u(0, t) = 0 \\ u(L, t) = 0 \end{cases} \quad \begin{cases} u(x, 0) = f(x) \\ u_t(x, 0) = 0 \end{cases} \quad \text{is given by}$$

(A) $u(x, t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right) \cos\left(\frac{n\pi t}{L}\right).$

(B) $u(x, t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi t}{L}\right) \cos\left(\frac{n\pi x}{L}\right).$

(C) $u(x, t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right).$

(D) $u(x, t) = \sum_{n=1}^{\infty} b_n \cos\left(\frac{n\pi x}{L}\right).$

Section C

Answer any **three** out of five questions.

Each question carries 4 marks.

14. Solve the logarithmic equation $\log(6y - 7) + \log y = \log 5$.
15. The smallest of 150 measurements is 5.18 in, and the largest is 7.44 in. Determine a suitable set of : (a) class intervals ; (b) class boundaries ; and (c) class marks that might be used in forming a frequency distribution of these measurements.
16. Four groups of students, consisting of 15, 20, 10, and 18 individuals, reported mean weights of 162, 148, 153, and 140 pounds (lb), respectively. Find the mean weight of all the students.
17. Prove that the sum of the deviations of $X_1, X_2, \dots, X_N$ from their mean $\bar{X}$ is equal to zero.
18. The bacterial count in a certain culture increased from 1000 to 4000 in 3 days. What was the average percentage increase per day ?

(3 × 4 = 12 marks)

Section D

Answer any **two** out of three questions.

Each question carries 6 marks.

19. The final grades in mathematics of 80 students at State University are recorded in the accompanying table :

68	84	75	82	68	90	62	88	76	93
73	79	88	73	60	93	71	59	85	75
61	65	75	87	74	62	95	78	63	72
66	78	82	75	94	77	69	74	68	60
96	78	89	61	75	95	60	79	83	71
79	62	67	97	78	85	76	65	71	75
65	80	73	57	88	78	62	76	53	74
86	67	73	81	72	63	76	75	85	77

With reference to this table, find :

- (a) The highest grade.
- (b) The lowest grade.
- (c) The range.

- (d) The grades of the five highest-ranking students.
- (e) The grades of the five lowest-ranking students.
- (f) The grade of the student ranking tenth highest.
- (g) The number of students who received grades of 75 or higher.
- (h) The number of students who received grades below 85.
- (i) The percentage of students who received grades higher than 65 but not higher than 85.
- (j) The grades that did not appear at all.

20. Find : (a) the quartiles Q_1 , Q_2 , and Q_3 and (b) the deciles D_1 , D_2 , ..., D_9 for the wages of the 65 employees at the P&R Company.

Wages	Frequency
250.00 - 259.99	8
260.00 - 269.99	10
270.00 - 279.99	16
280.00 - 289.99	14
290.00 - 299.99	10
300.00 - 309.99	5
310.00 - 319.99	2
Total = 65	

21. Find the moment co-efficient of skewness, a_3 for the height distribution of students at XYZ University :

Height (in)	Number of Students
60 - 62	5
63 - 65	18
66 - 68	42
69 - 71	27
72 - 74	8
Total = 100	

(2 × 6 = 12 marks)

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, NOVEMBER 2021**(CUCBCSS—UG)****Mathematics****MAT 5B 08—DIFFERENTIAL EQUATIONS****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 30 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
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MAT 5B 08—DIFFERENTIAL EQUATIONS

(Multiple Choice Questions for SDE Candidates)

1. The degree of the differential equation $(y')^{3/2} = y''$ is ?
- (A) 0. (B) 1.
(C) 2. (D) None of these.
2. The integral curves of the differential equation $y' = 1$ are ?
- (A) $y = x + c$. (B) $y = x^2 + c$.
(C) $y = x^3 + c$. (D) $y = x + 1$.
3. An integrating factor of the differential equation $ty' + 2y = 4t^2$ is ?
- (A) t^3 . (B) t^4 .
(C) t^2 . (D) None of these.
4. A homogeneous differential equation $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ can be converted to a variable separable equation using a transformation :
- (A) $y = vx$. (B) $y^2 = vx$.
(C) $y = vx^2$. (D) $y = v^2x$.
5. The differential equation $(6xy^2 + 4x^3y)dx + (6x^2y + x^4 + e^y)dy = 0$ is ?
- (A) A separable equation. (B) A linear equation.
(C) An exact equation. (D) A homogeneous linear equation.
6. An integrating factor of the differential equation $3(x^2 + y^2)dx + x(x^2 + 3y^2 + 6y)dy = 0$ is :
- (A) e^{-y} . (B) e^{2y} .
(C) e^y . (D) e^{-2y} .
7. The solution of the differential equation $y' = y^2$, $y(0) = 1$ exists in the region :
- (A) $(0, \infty)$. (B) $(-\infty, 0)$.
(C) $(-\infty, 1)$. (D) $(-\infty, \infty)$.

8. If $\frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$ is a function of y only, then an integrating factor of the differential equation $Mdx + Ndy = 0$ is :

(A) $\mu(x) = \exp \left[\int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy \right]$. (B) $\mu(x) = \exp \left[\int \frac{1}{M} \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y} \right) dy \right]$.

(C) $\mu(x) = \int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy$. (D) $\mu(x) = \int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy$.

9. An integrating factor of the differential equation $\frac{dx}{dy} + P(x) = Q(x)$ is :

(A) $e^{\int p dx}$. (B) $e^{-\int p dx}$.
(C) $e^{\int p^2 dx}$. (D) $e^{\int (P+Q) dx}$.

10. An integrating factor of the differential equation $\frac{dx}{dy} + Px = Q$ where P and Q are functions of y alone is :

(A) $e^{\int p dy}$. (B) $e^{-\int p dy}$.
(C) $e^{\int p dx}$. (D) $e^{\int p^2 dy}$.

11. The initial value problem $y' = y^y$, $y(0) = 0$, $t \geq 0$:

(A) A unique solution. (B) Infinitely many solutions.
(C) No solution. (D) Two solutions.

12. The general solution of the differential equation $3(x^2 + y^2)dx + x(x^2 + 3y^2 + 6y)dy = 0$ is :

(A) $x^3 e^{-y} + 3xy^2 e^y = c$. (B) $x^3 e^y + 3xy^2 e^y = c$.
(C) $x^2 e^y + 3x^2 y^2 e^y = c$. (D) $xe^y + ye^y = c$.

13. If $y_1(x)$ and $y_2(x)$ are two linearly independent solutions of the linear differential equation $a_0(x)y'' + a_1(x)y' + a_2(x)y = 0$ then :

(A) $y_1(x)y_2(x)$ is also a solution. (B) $y_1(x) + y_2(x)$ is also a solution.
(C) $y_1(x)/y_2(x)$ is also a solution. (D) $y_1^2(x) + y_2^2(x)$ is also a solution.

14. Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions of the differential equation $a_0(x)y'' + a_1(x)y' + a_2(x)y = 0$ then the Wronskian $w(y_1, y_2)$ is :
- (A) 1. (B) 0.
(C) 2. (D) -1.
15. The general solution of the differential equation $(D^2 - 4D + 4)y = 0$ is :
- (A) $(c_0 + c_1x)e^{2x}$. (B) $(c_0 - c_1x)e^{2x}$.
(C) $c_1e^x c_2e^{-2x}$. (D) None of these.
16. The Laplace transform of e^{at} is :
- (A) $\frac{1}{s-a}$. (B) $\frac{1}{s+a}$.
(C) $\frac{1}{s^2-a^2}$. (D) $\frac{1}{s^2+a^2}$.
17. The Laplace transform of $\cos at$ is :
- (A) $\frac{s}{s^2+a^2}$. (B) $\frac{1}{s^2+a^2}$.
(C) $\frac{1}{s^2-a^2}$. (D) $\frac{s}{s^2-a^2}$.
18. If $L\{f(t)\} = F(s)$, then $L\{f(at)\} =$
- (A) $\frac{1}{a} f(s/a)$. (B) $F(s/a)$.
(C) $F(a/s)$. (D) $F(s)$.
19. The Laplace transform of the delta function is :
- (A) e^{-as} . (B) e^{as} .
(C) $e^{as/s}$. (D) $e^{-as/s}$.
20. $\int_0^\infty \frac{\sin t}{t} dt =$
- (A) $\frac{\pi}{4}$. (B) $\frac{\pi}{8}$.
(C) $\frac{\pi}{2}$. (D) None of these.

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, NOVEMBER 2021

(CUCBCSS-UG)

Mathematics

MAT 5B 08—DIFFERENTIAL EQUATIONS

Time : Three Hours

Maximum : 120 Marks

Part A*Answer all questions.**Each question carries 1 mark.*

1. Prove that the product of two odd functions is an even function.
2. Prove $L^{-1}\{1\}$.
3. Write down the differential equation whose solution is $y = c_1 e^{5t} + c_2 e^{-2t}$.
4. Evaluate $W[e^{\mu \cos \lambda t}, e^{\mu \sin \lambda t}]$.
5. Compute $L\{t^2 e^{\lambda t}\}$.
6. Find the integrating factor of $(x-2)(x+1) \frac{dy}{dx} + 3y = x$.
7. Solve the system $\frac{dy}{dt} - x = 0, \frac{dx}{dt} - y = 0$.
8. Find the fundamental solutions of $y'' + 25y = t^{-1/2}$.
9. Find the value of b_n in the Fourier sine series expansion of 2π -periodic function $f(x) = -x, x \in [-\pi, \pi]$.
10. Write one dimensional heat equation with all the assumptions involved.
11. What do you mean by an exact differential equation? Give an example.
12. Find the complementary function corresponding to $y'' - 2y' + 2y = t$.

(12 × 1 = 12 marks)

Part B*Answer any ten questions.**Each question carries 4 marks.*

13. Convert $y'' + 2y' = 0$ into a system of first order equations.

Turn over

- 14 Find the Fourier cosine series for the 2π -periodic function $f(x) = x, x \in [-\pi, \pi]$.
- 15 Find the integrating factor for $(2x + 3y)dx + (2x - 3y)dy = 0$.
- 16 Find the inverse Laplace transform of $\log((s-a)/(s-b))$.
- 17 Define unit step function and find its Laplace transform.
- 18 Solve $t^2x'' - 2tx' - 3x = 0$.
- 19 Write the existence and uniqueness theorem for first order differential equations with the assumptions involved therein.
- 20 Show that the inverse Laplace transform is linear.
- 21 State Abel's theorem.
- 22 Solve $\frac{dy}{dx} = (3x + 2y + 1)^2$.
- 23 Evaluate $L\{te^t \cos 2t\}$.
- 24 Find the second order p.d.e. for which $y = \phi(x+at) + \psi(x-at)$ is a solution.
- 25 Solve the system : $\frac{dy}{dt} = x - y, \frac{dx}{dt} = x + y$.
- 26 Solve $y' - 2y = 0$ using Laplace transform.

(10 × 4 = 40 marks)

Part C*Answer any six questions.**Each question carries 7 marks.*

- 27 Express the function $f(t) = \begin{cases} t \sin t, & \text{if } 0 \leq t < \pi/2 \\ \cos t, & \text{if } \pi/2 \leq t < \pi \\ 0, & \text{elsewhere} \end{cases}$ in terms of combination of unit step functions and hence find its Laplace transform.
- 28 Evaluate the Laplace inverse transforms of $4 - \cot^{-1}(s/a)$ and $\frac{1}{(s^2 - 5s + 6)^2}$.
- 29 Find the solution by checking the exactness of $(3y^2 - 2xy + 2)dx + (6xy - x^2 + y^2)dy = 0$.
- 30 State the conditions for the existence of Laplace transform of a function $f(t)$ and prove the same.